AEGIS AI v2

Decentralized AI Trust, Provenance, & Escrow Platform on Cardano

TEAM SELECLAW

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THE AI DATA INTEGRITY CRISIS

Three Core Challenges Blocking Enterprise AI Adoption:

1. Untrusted Lineage

AI models are trained on datasets with zero cryptographic proof of consent, leaving companies open to severe copyright infringement claims.

2. Centralized Audits

Current verification systems use black-box centralized auditors. Indices like compliance and training readiness remain completely unverified.

3. Inefficient Payments

Data creators face high intermediary fee margins and slow transaction settlements. There is no automated revenue split on secondary markets.

TECHNICAL ARCHITECTURE

Ecosystem Core Components:

- **Lucid Evolution Integration**
  Interfaces directly with Cardano wallets (Lace, Nami) using CIP-30 standards to build, sign, and submit on-chain ledger actions.

- **Aiken Escrow Contracts**
  Plutus smart contracts validating locked Lovelace, splitting revenue (95% to seller, 5% to treasury), and minting License NFTs.

- **Pinata IPFS Metadata Vault**
  Immutable, decentralized storage pinning dataset registries, scanning logs, and model metadata verification schemas.

TECH STACK OVERVIEW

FRONTEND:
Next.js 15 (Tailwind & custom CSS)

SMART CONTRACTS:
Aiken Plutus Validator

WEB3 CLIENT:
Lucid Evolution (WASM)

DECENTRALIZED IPFS:
Pinata Gateway API

NETWORKS:
Cardano Preprod Testnet

CORE PLATFORM INNOVATIONS

9-Node Provenance Lineage

Auditable trajectories mapping datasets from the medical provider to the model checkpoint and royalties distribution on the ledger.

Triple-Index Scoring Engine

Side-by-side verification of Dataset Trust, AI Readiness, and regulatory compliance (HIPAA/GDPR) calculated in the upload scan.

Secure AI Sandbox Workspace

Containerized training workspace monitoring telemetry (GPU Temp, CPU Load, VRAM) and exporting weights directly to on-chain checkpoints.

Dependency Model Registry

An audit registry mapping registered ML weights back to their exact verified dataset roots, preventing model spoofing.

SYSTEM STEP-BY-STEP WORKFLOW

1

AUDIT & UPLOAD

Creator uploads dataset categories. System scans for security formatting, calculating trust and readiness scores.

2

IPFS PIN & NFT MINT

Metadata is pinned to Pinata IPFS. Cardano Preprod ledger mints the Dataset Certificate NFT to establish provenance.

3

SMART ESCROW LOCK

Buyer purchases dataset access. Lucid locks ADA in the Plutus Escrow script, auto-minting a License NFT directly to the buyer.

4

TRAIN & EXPORT WEIGHT

Buyer mounts dataset in Secure Workspace. Model completes training, and user exports weights checkpoint NFT on-chain.

5

REGISTRY LINKAGE

Finalized model weights are registered, permanently cataloging the model linked to its verified training dataset dependency.

SMART ESCROW & PLATFORM ECONOMICS

P2P ESCROW DISTRIBUTION

Aegis AI eliminates centralized brokerage cuts using direct ledger settlement rules:

- Escrow datum stores creator address, metadata URL, and purchase Lovelace values.
- Payments route directly peer-to-peer on Cardano Preprod Testnet.
- License NFT is minted and transferred atomically inside a single transaction bundle.

5% TREASURY SPLIT MODEL

Smart contract automatically enforces a 5% split of transactions into the treasury pool:

- 95% of dataset purchase fees go directly to the dataset creator.
- 5% platform commission is routed to the Community Governance Treasury pool.
- Treasury funds are unlocked dynamically via stake-weighted community votes.

LIVE API & KEY CONFIGURATION

Verifiable Blockchain & Storage Integrity API setup:

Blockfrost Preprod Provider API

Acts as our gateway connection directly to the Cardano Preprod ledger. Used by Lucid Evolution for script state evaluation, datum reads, UTxO querying, and transaction execution.

Pinata IPFS Gateway JWT

Provides secure authentication to pin datasets and JSON schemas to decentralized nodes, generating immutable, tamper-proof IPFS CIDs.

Plutus Escrow Validator Hash

The unique hash derived from our compiled Aiken validator. Configured inside environment settings to identify the exact ledger locking address.